

The Wave, the River, and the Self: A Boundary-Condition Theory of Identity

Abstract

What constitutes identity?

The traditional answers have oscillated between two extremes. Materialist theories locate identity in matter itself: we are the atoms, molecules, and physical components that compose us. Dualist theories locate identity in a separate immaterial substance: soul, mind, or spirit.

Both approaches encounter serious difficulties.

The materialist view struggles to explain persistence through continuous material replacement. The dualist view introduces entities that remain difficult to formalize within the natural sciences.

This paper proposes a third alternative.

Identity is neither the matter itself nor a separate substance. Identity is the persistence of a structural pattern maintained by boundary conditions and constraints that organize continuously renewed material flows.

The wave, the river, and the living organism all reveal the same principle: persistence does not require material permanence. It requires structural continuity.

1. The Problem of Identity

A simple question has troubled philosophers for centuries:

If all the material components of a system change, why do we still regard it as the same thing?

The question appears in many forms:

- The Ship of Theseus.
- Personal identity through time.

- Biological continuity.
- The persistence of institutions and cultures.

The difficulty emerges because our language naturally assumes that identity belongs to substances.

We assume that an object remains itself because some material core remains unchanged.

Yet many familiar systems do not behave this way.

They persist despite continuous material replacement.

To understand why, we must begin not with people, but with waves.

2. The Wave

Consider an ocean wave.

At first glance, the wave appears to be a moving mass of water.

Closer examination reveals something different.

If we observe a fixed location in the ocean, the water molecules do not travel across the sea with the wave.

Instead, they oscillate locally.

They rise.

They fall.

They move in small circular or elliptical trajectories.

The disturbance travels.

The water largely does not.

The identity of the wave therefore cannot reside in the individual molecules composing it.

The molecules are constantly changing.

The wave persists.

What remains is the pattern.

In a precise sense, the wave is closer to a mathematical solution than to a material object.

Its identity resides in structure rather than substance.

3. The River

The same principle appears in rivers.

The water present in a river today is not the water that was present yesterday.

Every droplet is replaced.

Yet we continue to speak of the same river.

Why?

Because the continuity of the river is not provided by the water.

It is provided by the organization of the flow.

The geometry of the riverbed constrains possible trajectories.

The banks channel motion.

The depth influences velocity.

The hydraulic radius determines how energy is distributed throughout the system.

The river remains identifiable because the structure governing the flow remains sufficiently stable.

The identity of the river resides more in its boundaries than in its contents.

4. Engineering and Boundary Conditions

Engineering provides an even clearer example.

Consider a beam.

The governing differential equation may remain exactly the same.

Yet the behavior of the beam changes dramatically depending on its boundary conditions.

A fixed beam behaves differently from a simply supported beam.

A cantilever behaves differently from both.

Nothing changes in the local physics.

The governing equation remains identical.

What changes are the constraints.

The solution depends not only on the equation itself but on the conditions imposed at its boundaries.

Identity emerges from the interaction between dynamics and constraint.

The restriction is as important as the mechanism.

5. The Living Body

The human body presents the same phenomenon on a larger scale.

Atoms enter and leave continuously.

Proteins are synthesized and degraded.

Cells die and are replaced.

Many components of the body are renewed repeatedly throughout life.

Yet personal identity persists.

The material substrate changes.

The individual remains.

This observation creates a challenge for strong materialism.

If identity were identical to matter, then continuous material replacement should imply continuous identity replacement.

But this is not what we experience.

A structural perspective offers a different explanation.

The organism persists because the organizational pattern persists.

Matter flows through the structure.

The structure itself maintains continuity.

The body resembles a river more than a stone.

6. Selection and Boundary Conditions

The argument becomes even more interesting when we consider systems whose constraints can themselves be modified.

Consider a simple physiological example.

A person may choose to ingest calcium or magnesium.

The intestinal absorption mechanisms belong to the body's internal dynamics.

Yet the choice of which substance enters the system originates outside those dynamics.

The internal equations remain unchanged.

The boundary conditions change.

The resulting physiological trajectory changes accordingly.

The same principle appears in artificial intelligence.

A language model may remain completely frozen.

Its weights do not change.

Its architecture does not change.

Yet a different system prompt can produce radically different outputs.

The internal dynamics remain constant.

The boundary conditions are altered.

Again, structure changes behavior without modifying the underlying mechanism.

7. The Limits of Substance-Based Identity

These examples suggest that our ordinary language may be misleading.

Words such as "thing," "object," and "entity" imply material permanence.

Yet many persistent systems lack such permanence.

A wave is not its water.

A river is not its droplets.

A living organism is not its atoms.

Their continuity emerges from organizational persistence.

The traditional search for identity within matter may therefore be misguided.

The relevant question is not:

"What substance remains?"

But rather:

"What structure persists?"

8. A Structural Theory of Identity

The examples discussed throughout this paper suggest a common principle.

Identity is not a property of matter.

Identity is a property of organized persistence.

Matter provides implementation.

Structure provides continuity.

A system remains itself when the constraints that organize its dynamics remain sufficiently stable despite ongoing material replacement.

Under this view, persistence is fundamentally structural.

Objects are not primarily collections of material components.

They are stable patterns sustained through constrained flows.

The apparent permanence of matter is secondary.

The continuity of organization is primary.

9. Conclusion

The wave is not its water.

The river is not its droplets.

The self is not its atoms.

In each case, persistence emerges not from material permanence but from the continuity of structure.

The central insight is simple:

Identity resides not in what flows through a system, but in the constraints that organize that flow.

Matter changes.

Structure persists.

And it is this persistence of structure, rather than persistence of substance, that provides the most coherent account of identity across physics, engineering, biology, and human experience.

Perhaps the self is neither a thing nor a soul in the traditional sense.

Perhaps it is something closer to a wave:

a stable pattern sustained through a continuously changing material world.

Appendix - The Wittgensteinian Problem of Identity

Why the question "what is the self?" is wrongly formulated

Philosophical Note — Claudio Bresciano

1. The trap of language

Wittgenstein identified a fundamental problem in philosophy: many of our questions are **badly formulated from the start** because language forces us to use grammars that do not correspond to the phenomenon.

The classic question:

■ *"What is the self?"*

This question presupposes that the self is **something** — a substance, a matter, a tangible thing. The grammar of "what" demands a noun as answer. Language forces us to search for a **thing**.

But the wave described by:

$$A \sin(kx - \omega t)$$

is not a thing. It is not matter. It is not tangible. It does not occupy a place in the sense that a stone occupies a place.

The wave is a **structure**. A **form**. A **pattern** of reality.

2. The wave equation as ontological metaphor

Consider the equation:

$$\Psi(x, t) = A \sin(kx - \omega t)$$

Where is the wave?

- It is not in the water (molecules oscillate locally, they do not travel)
- It is not in the air (the medium is contingent)
- It is not in the number A (amplitude is a parameter, not the wave)
- It is not in k or ω (they are constants of the solution)

The wave is in the **relation**. In the pattern that emerges when medium, boundary conditions, and dynamics couple.

The wave is **real** — it produces measurable effects, transports energy, modifies matter — but it is not **substantial**.

This is crucial: **reality** $\neq$ **substantiality**.

3. Descartes' error

Descartes formulated:

■ *"Cogito, ergo sum" — I think, therefore I am.*

But the "sum" (I am) led him to conclude that there is a **res cogitans** — a thinking substance. The Latin language, with its grammar of nouns and verbs, forced him to reify thought.

Wittgenstein would have said: the error is not in the conclusion, but in the question. "I think" does not imply a thinking agent, just as "it rains" does not imply a "rainer".

The self is not the **substance** that thinks. The self is the **pattern** that emerges when thinking occurs.

4. Why "what?" is the wrong question

QUESTION	PRESUPPOSITION	PROBLEM
What is the self?	It is a thing/substance	The self is not a thing
Who am I?	It is an agent/subject	The self is not a separate agent
Where is the self?	It has spatial location	The self is not localized

QUESTION	PRESUPPOSITION	PROBLEM
When did I begin to be myself?	It has a temporal beginning	The self is continuous pattern, not event

The appropriate question is not "what?" but "**how?**":

"How does identity emerge?"

Or better yet:

"What conditions maintain the continuity of the pattern?"

5. The river and the right question

Heraclitus said you cannot step into the same river twice. This is true if the river is its drops. But it is false if the river is its **flow pattern**.

The question "what is the river?" has two incompatible answers:

- **Substantial answer:** the river is the water drops → the river constantly changes → there is no identity
- **Structural answer:** the river is the flow pattern organized by the bed → the river persists → there is identity

The first answer generates the problem of identity. The second dissolves it.

The problem is not philosophical. It is **linguistic**. We use a language designed for substantial things to describe structural patterns.

6. Implications for the problem of the soul

If the self is a pattern, what about the soul?

The dualist tradition asks:

"Is the soul a substance separate from the body?"

But this question presupposes that the soul, if it exists, must be **substantial**. The grammar of "is..." demands an ontology of things.

If we reformulate:

■ *"Is the soul the pattern that organizes the flow of experience?"*

The answer no longer requires dualism. The soul is **structural**, not substantial. It is the form of the wave, not the water. It is the bed of the river, not the drops.

This is not materialism (the pattern is not reducible to matter). It is not dualism (the pattern is not a separate substance). It is **ontological structuralism**: fundamental reality is pattern, not substance.

7. TNA and the dissolution of the problem

The Theory of Axiomatic Necessity formalizes this intuition:

- **N0**: the operational dynamics (the drops, the atoms, the processes)
- **N1**: the boundary structure that organizes the dynamics (the bed, the pattern, the form)
- **Identity**: the continuity of the N1-N0 coupling through time

The self is not N0 (we are not our atoms). The self is not pure N1 (we are not a disembodied form). The self is **the river**: the flow pattern that emerges from the coupling.

The question "what is the self?" dissolves because there is no separable "what". There is a **how**: the mode in which N1 organizes N0 into a recognizable trajectory.

8. Conclusion: beyond the "what"

Wittgenstein wrote in the *Tractatus*:

■ *"Whereof one cannot speak, thereof one must be silent."*

But perhaps he should have added:

■ *"Whereof one cannot ask 'what', one must ask 'how'."*

The self is not a thing to be inquired about. It is a pattern that maintains itself when matter flows. Identity is not substance. It is **structural continuity**.

The wave is not water. The river is not drops. The self is not atoms.

They are **forms**. And forms are real.